

# Dexter Fernandes

Bristol • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns

## Computer Vision Engineer

Computer Vision Engineer with experience building and shipping deep learning models for real-world image analysis, from data preprocessing and training pipelines through deployment, production monitoring, and post-deployment validation. Strong background in Python, PyTorch, TensorFlow, OpenCV, ONNX, and Docker, with hands-on experience delivering low-latency, reliable CV systems in demanding production environments. Built end-to-end workflows spanning data curation, augmentation, training, optimisation, observability, and user-facing applications, with a strong focus on robustness, model performance, and practical business impact. Comfortable translating research and prototyping into production-ready computer vision systems for cross-functional teams and client-facing use cases.

## WORK EXPERIENCE

### Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

#### AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation.
- Built multimodal data ingestion, cleansing, training, and validation pipelines using Python, Pandas, Apache Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours, handling upto 3.6 million records.
- Built synthetic image generation and augmentation procedures with OpenCV and SciPy to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Applied classical image-processing techniques to detect mud and sand splotches on camera lenses, reducing false positives by 13% and improving system reliability in harsh industrial conditions.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.

### LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

#### Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% on edge devices.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Optimised model architectures using quantisation-aware training, structured pruning, layer fusion, FP16/INT8 deployment, and targeted profiling, reducing model size by 50% while meeting accuracy, latency, and memory constraints.
- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% while accelerating iteration cycles on large real-world datasets.
- Owned end-to-end production ML pipelines spanning data ingestion, labelling, training, optimisation, deployment, monitoring, and field validation for industrial safety systems, improving deployment reliability and release consistency.

- Built a low-latency, high-reliability computer vision PoC for a zinc ore refining plant in South Africa, and developed a user-facing hazard detection application for an Oman-based client.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Mentored junior engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.

## PROJECTS

### Modern Edge CV Benchmarking

Mar 2026

- Benchmarked modern computer vision models including YOLOv8, YOLOv10, and RT-DETR on a shared validation subset, comparing mAP, inference time, and model size to understand trade-offs for low-latency deployment.
- Exported models to ONNX and evaluated edge deployment paths using OpenVINO, strengthening practical understanding of optimisation and model selection for production-style CV systems.
- Implemented and evaluated recent edge and foundation-model approaches, including Grounding DINO, MobileSAM, Depth Anything v2, and EfficientViT, to compare deployment trade-offs across accuracy, latency, and model size.

### End-to-End Damage Severity & Claim Assessment

Nov 2025

- Developed an end-to-end automotive claims assessment pipeline combining computer vision–based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

## EDUCATION

### Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

#### Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation before real-world deployment.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation, and debugged tf trees and frame connectivity.

### Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, perception systems, and robotics-adjacent engineering.

## CERTIFICATIONS

### Building RAG Agents with LLMs

Mar 2026 - Apr 2026

Nvidia

## AWARDS & SCHOLARSHIPS

### Best Speaker award

Jan 2020

ABC Toastmasters club

## SKILLS

- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, YOLO
- **Computer Vision Models:** ByteTrack, Depth Anything v2, EfficientDet, EfficientViT, Faster R-CNN, Grounding DINO, MobileSAM, RT-DETR, SSD MobileNet V2, YOLOv10, YOLOv5, YOLOv8
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, OpenCV, Pandas, PIL, PostgreSQL, Python, Qt, scikit-image, SciPy
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering